Apex Precision Manufacturing Pvt. Ltd.

Equipment Maintenance & Troubleshooting Manual

CONTROLLED SAMPLE DOCUMENT FOR RAG CHATBOT TESTING

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Version	1.0 – Sample	**Effective date**	30 October 2026
Document owner	Maintenance Department	**Review status**	Training sample
Classification	Internal – Fictional	**Approval**	Not approved for real operations

IMPORTANT NOTICE

This document is fictional sample content created solely to develop and test a Retrieval-Augmented Generation (RAG) chatbot. It is not an approved company procedure, machine-specific instruction, legal policy, or substitute for manufacturer documentation, site risk assessments, qualified supervision, or applicable laws. Do not use it to operate, repair, inspect, or service real equipment.

Purpose

Describe a safe, high-level process for reporting equipment faults, conducting authorized inspections, and documenting maintenance activity.

Document control

The document owner is responsible for review and controlled distribution in a real organization. For this training dataset, version 1.0 is fixed and all names, policies, workflows, and examples are fictional.

1. Scope and authorization

• This sample guide covers general CNC machining-center maintenance coordination and basic fault reporting.

• Only trained and authorized maintenance personnel may diagnose or service equipment.

• Electrical, hydraulic, pneumatic, stored-energy, and mechanical work must follow the site's approved isolation and lockout/tagout procedure and applicable law.

2. Fault reporting workflow

• Stop work and make the area safe if abnormal operation, smoke, fluid leakage, unusual noise, overheating, or a safety fault is observed.

• Notify the shift supervisor and maintenance team; provide the machine ID, time, alarm code, operating condition, and recent events.

• Do not repeatedly reset alarms or restart equipment to keep production moving.

• Mark equipment out of service where required by site procedure until an authorized person releases it.

3. Troubleshooting information to collect

• Record the exact alarm message and code as displayed; do not guess at the cause.

• Note whether the fault occurs at startup, during a cycle, at tool change, or during idle operation.

• Record relevant readings only when the operator interface or approved checklist instructs the employee to do so.

• Check the equipment history and manufacturer documentation before deciding on corrective action.

4. Common symptom categories

• Over-temperature indication: stop and escalate; an authorized technician should review cooling, ventilation, lubrication, load, and sensor information against manufacturer limits.

• Abnormal vibration or noise: stop the process safely and request inspection of workholding, tooling, bearings, and machine condition by qualified personnel.

• Coolant or lubrication warning: follow the machine-specific alarm response and approved fluid-handling procedure; do not operate against an active protective warning.

• Repeated electrical or control alarm: preserve the alarm details and refer to qualified electrical/controls personnel.

5. Preventive maintenance planning

• Use the approved asset maintenance schedule and manufacturer recommendations for intervals and tasks.

• Record completed inspections, replaced parts, consumables, calibration status, and outstanding defects.

• Do not substitute generic intervals in this document for machine-specific service intervals.

• Return equipment to production only after required checks and authorized release are documented.

6. Maintenance records

• Work order; fault report; isolation record; inspection checklist; parts usage record; post-maintenance verification and release.

Revision history

Version	Date	Description	Prepared by
1.0	30-Oct-2026	Initial fictional sample for RAG testing	Project team